

Hanyi Zhao

Email: 220320829@stu.hit.edu.cn

GitHub: <https://github.com/Zzzhy0919>

Research Interests

- My research interests mainly focus on the field of robotics, with a focus on robot control systems and artificial intelligence (AI), particularly in the realm of computer vision. By combining AI with robotics, I hope to explore ways for robots to interact and collaborate smoothly with humans in complex environments, in order to achieve more intelligent robot systems.

Education

- **Harbin Institute of Technology(Shenzhen Campus)** *Sep 2022 - present*
 - Degree: B.E. in Automation
 - GPA: 91.758/100
 - National Scholarship, First-Class Academic Scholarship (two times)

Experience

- **Research Intern with Prof. Xueqian Wang** *Oct 2023 - Feb 2025*
Center for Artificial Intelligence and Robotics, Tsinghua SIGS, Shenzhen
 - Designed a novel pipeline that learns basic skills from long demonstrations and composes learned basic skills to generalize to unseen multi-step cloth manipulation tasks.
 - Proposed an autonomous basic skill discovery method for cloth manipulation, where an LLM is used to provide commonsense knowledge.
 - Conducted extensive experiments to validate the effectiveness and generalization of our proposed pipeline in both simulation and real-world environments.
 - Project website: <https://sites.google.com/view/gen-cloth>

Projects

- **RoboMaster Drone Design and Optimization** *Nov 2022 - Aug 2023*
 - Designed a lightweight but high-performance unmanned aerial vehicle, including gimbal, frame, propeller protective cover, etc.
 - Used 3D printing, laser cutting, precision carving machine and other technologies to complete the construction of the whole drone.
- **Intelligent Car Using Computer Vision Design and Optimization** *Dec 2023 - Jul 2024*
 - Designed and built a small car that can efficiently complete various tasks and is as lightweight as possible.
 - Designed a set of algorithms, including chassis control, serial communication modules, etc., to enable the car to adapt to different task environments and complete specified tasks with the highest possible accuracy and speed.
 - Performed digital image processing and image classification on images captured by different cameras, enabling the car to complete tracking and image classification tasks.

Skills

- **Programming Languages:** Python, C++, C, Matlab
- **Frameworks & Tools:** Linux, ROS, PyTorch, Windows
- **Interests:** Photography, Guzheng, Table tennis
- **Languages:** English (fluent), Chinese (native)

Awards & Certifications

- H Award of American College Student Mathematical Modeling Competition *May 2024*
- First Prize of Guangdong Province in the Blue Bridge Cup National Software and Information Technology Professional Talent Competition *Jun 2024*
- Second Prize in the South China Division of the National College Student Intelligent Car Competition *Jul 2024*